Saliency and Observer Disagreement Prediction with a Center-Biased Deep Encoder-Decoder

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***Abstract*—Saliency prediction aims to estimate where humans look within a scene, and supports a range of downstream vision tasks such as object recognition, tracking and image compression. We propose a redesign of the Deep Visual Attention Prediction model, built around a pre-trained ResNet34 encoder, U-Net-style skip connections, a learnable center-bias prior with Gaussian output smoothing, and a loss that combines the Kullback-Leibler divergence with the Linear Correlation Coefficient. With a markedly lighter architectur with respect to the more recent transformer-based , the model surpasses the original on CC and SIM on MIT1003 without any fine-tuning on that dataset. Observing that averaging over observers also discards whether they agreed, we introduce inter-observer disagreement as a second prediction target, quantified through the Jensen-Shannon divergence across per-observer fixation densities and corrected by a permutation null that removes the upward bias arising from small samples. At almost no additional cost, the model can therefore estimate how strongly observers are expected to agree on a given image, which in turn helps to tell whether a poor saliency prediction reflects the inherent ambiguity of the image or a limitation of the model itself.**

***Index Terms*—Saliency Prediction, Encoder-Decoder, Deep Supervised Learning, Inter-Observer Agreement, Convolutional Neural Networks, Eye Tracking**

I. Introduction

Visual saliency detection simulates the ability of the Human Visual System (HVS) to select and focus on the most relevant parts of a scene. This modelling is important because it enables a variety of tasks, such as object recognition, visual tracking, object segmentation or image compression.

It is usually costly and difficult to obtain human eye-tracking data, which is why the task of saliency prediction has been widely studied during the years. Many traditional handcrafted approaches and deep learning models have been proposed to predict saliency maps from images, that are heat maps where the brightness of each pixel represents the probability that a human will focus on it. Deep learning models achieve state-of-the-art performance, in particular *Convolutional Neural Networks* (CNNs) have been widely used for this task, due to their ability to extract local features from images. For example, *Deep Visual Attention Prediction* [1] is an innovative approach that implements an encoder-decoder architecture based on CNNs: the encoder extracts hierarchical features and the multi-decoder fusion strategy combine them in order to consider both local and global information about the image. We used [1] as the reference for our work, proposing and analyzing an improved version of their model.

However, most of these models are trained and evaluated on the *aggregate* saliency map obtained by averaging the fixations of many observers on the same image, and performance on this aggregate target is starting to saturate, with reported improvements of a few hundredths of a point often falling within seed-to-seed variance.

Moreover, the aggregate map itself hides information: two images can produce nearly identical average maps even though, in one, every observer looks at the same region, while in the other observers split evenly between two distant ones, and no model trained only on the aggregate map can tell these cases apart.

The main contributions of this work are the following:

- Propose a series of modifications to the *Deep Visual Attention Prediction*'s model for the saliency prediction task, to obtain better results, while maintaining the simplicity and clarity of the reference model.
- Propose a loss function that combines the *Kullback-Leibler divergence* and the *Linear Correlation Coefficient* to improve the performance of the model.
- Evaluate the proposed model on the MIT1003 dataset [2] and compare it with other state-of-the-art models, demonstrating competitive performances in terms of obtained results.
- Compare three encoder/fusion tiers – a single-branch ResNet, a multi-decoder fusion model and a transformer-backboned variant – to check whether the added architectural complexity still improves standard saliency metrics.
- Propose predicting *inter-observer disagreement* as a new target, computed from the individual eye-tracking records that standard saliency pipelines discard, and evaluate it with a dedicated protocol.

The remaining text is organized as follows. In Section II we review some state-of-the-art-models for saliency prediction, in Section III and IV we describe the proposed model and the datasets used for training and testing it. In Section V we provide a more detailed description of the network architecture and the approach used to train the model. In Section VI we present the metrics used to evaluate the obtained model and the experimental results. Finally, in Section VII we draw some conclusions and discuss future work.

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II. Related Work

Saliency prediction has been addressed with both hand-crafted and learned models, the latter now dominating thanks to the ability of CNNs to capture complex visual patterns. The approach we take as reference is [1], an encoder-decoder network with a multi-decoder fusion strategy: a VGG16 encoder [3] pre-trained on ImageNet [4] feeds three decoders branching from different intermediate representations, each producing an auxiliary saliency map under deep supervision, and the three maps are then fused into the final prediction. Its encoder is however comparatively dated, and no mechanism recovers the spatial detail lost during downsampling, which is what our redesign addresses.

Among purely convolutional models, [5] keeps a feed-forward design and inserts an ASPP module that aggregates multi-scale context through parallel dilated convolutions, reaching state-of-the-art accuracy on MIT300 while remaining lightweight.

More recent work introduces transformers to capture the long-range dependencies that convolutions encode poorly, even though they demand substantially more computation and data. For intance, [6] inserts transformer modules into a CNN encoder-decoder and trains on a linear combination of four saliency metrics, reaching state-of-the-art results on public benchmarks.

The individual eye-tracking records behind the aggregate maps have received far less attention. The MIT1003 dataset [2] provides the fixations of its 15 observers separately, but most work collapses them into a single average map and never uses them again. Where observer variability has been studied, it has served as a benchmarking device: [7] predicts each held-out subject from the others, using inter-subject consistency as an upper bound on what a generic model can achieve. This bound is a population-level quantity rather than a per-image value a model can output, and it assumes enough observers to be stable. We instead compute disagreement per image, correct it for small-sample noise, and treat it as a prediction target in its own right.

III. Processing Pipeline

The approach we propose is a redesign of the architecture described in [1] in terms of data preprocessing, learning procedure and network structure, incorporating skip connections and a modified encoder to enhance the performance.

We implement an architecture based on encoder-decoder structure. The encoder backbone in our setting is a ResNet34 network [8], from which three different intermediate representations are extracted and passed as input to a multi-depth decoder structure. The decoder is composed of three different decoders, which perform upsampling operations in order to predict three different saliency maps. Combined with the implementation as a skip-layer network, it allows to capture multi-scale information, since it was proven by the authors to be beneficial for capturing hierarchical saliency information from different levels of the HVS. In the end, the three obtained maps are then fused together in order to obtain the final prediction.

Skip connection are added between the encoder and the decoders in a U-net [9] fashion, in order to add to the decoder layers the information lost while performing the downsampling operations. The network architecture is shown in Fig. 1.

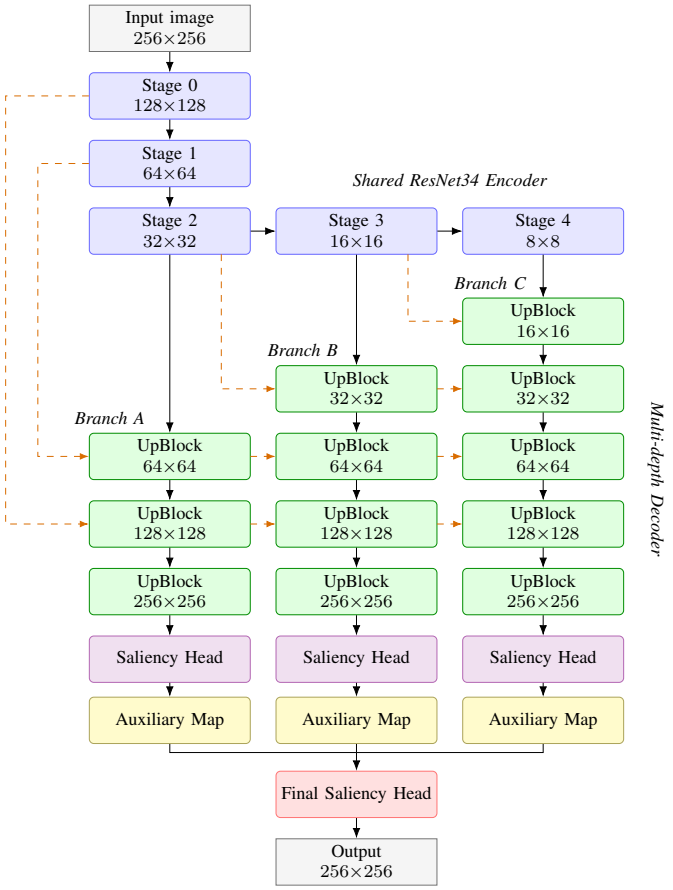


Fig. 1: Network architecture. Dashed arrows rappresent the skip connections.

The network is trained on the SALICON dataset using Adam optimizer and a deep supervised loss with a *Kullback-Leibler divergence* plus *Linear Correlation Coefficient* loss, in order to model the problem as a probability distribution task and optimizing at the same time the correlation between the ground truth distribution and the saliency map predicted by the model.

The pipeline described so far predicts the aggregate saliency map, and is trained end-to-end on SALICON. In parallel we build a second branch, which targets a quantity that the aggregate map cannot express by construction. Starting from the per-observer eye-tracking records of MIT1003, we reconstruct one fixation density per observer, measure how much these densities disagree with each other, and correct that measurement for the noise introduced by the small number of fixations each observer contributes. This produces a single number per image which is then used as the target of a second

lightweight head attached to the same encoder that feeds the saliency decoders. The two branches share their features and a single forward pass, but predict different quantities: the decoders predict *where* people look, while the second head predicts *how much people differ* in where they look. The overall flow is shown in Fig. 2.

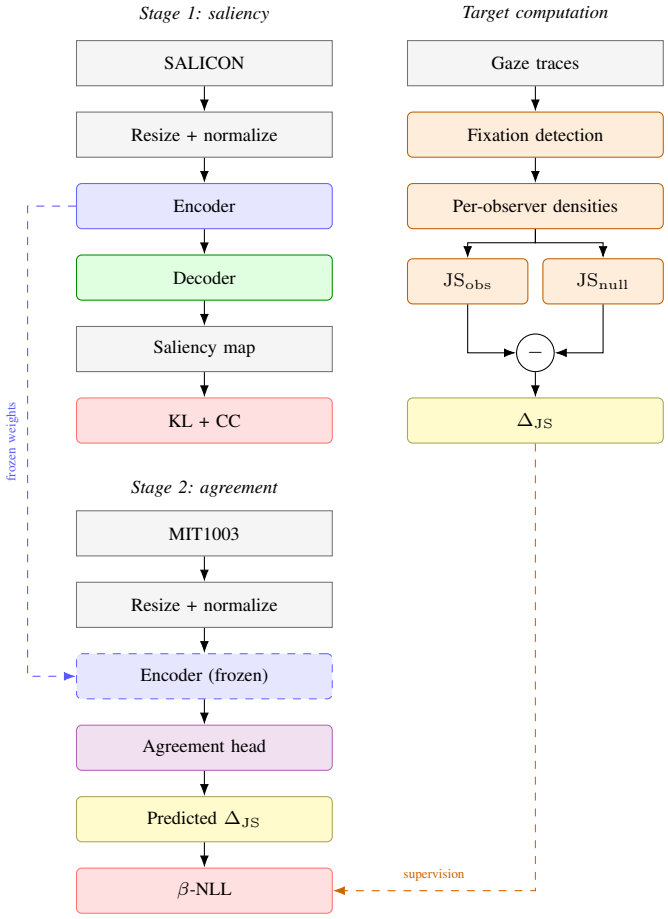


Fig. 2: Processing pipeline of the two branches.

IV. Dataset and Preprocessing

The network was trained and validated on the SALICON dataset [10], which contains $20,000$ images from the MS COCO dataset specifically labeled for the task of Visual Saliency Prediction. In this dataset, the images are divided into $10,000$ images for training, $5,000$ images for validation and $5,000$ images for testing, even if the ground truth for the latter set is not published by the authors.

The testing was performed with the MIT1003 dataset [2], which contains $1,003$ images labeled for the Saliency Prediction task, since for this dataset the ground truth saliency map is available.

All the training, validation and test images where preprocessed by resizing them to a standard 256×256 resolution and normalized by the mean and standard deviation values of the images in the ImageNet dataset, since the encoder backbones are pre-trained on that dataset. Moreover, the aspect ratio of the images is kept intact by adding padding to both the image and the saliency map on the shorter size, so as to not alter the proportions of the features of the images by forcing a squared aspect ratio.

Furthermore the training set is augmented with random horizontal flipping, applied stochastically at load time with probability $p = 0.5$. This does not increase the number of training samples per epoch, but exposes the model to both the original and mirrored composition of a given scene across different epochs, increasing the diversity of views seen during training.

Per-Observer Fixation Data

The agreement branch does not use the aggregate maps at all, but the raw eye-tracking records that MIT1003 distributes as one file per observer and image. These files store the gaze trace as sampled by the eye tracker rather than a list of fixations: a 3 s trial produces several hundred samples, most of them repeated measurements taken while the eye dwells in the same place. Taking each sample as a fixation would yield around 700 fixations per observer instead of the expected ten, making every observer's density look nearly identical and destroying the signal this branch is meant to measure. We therefore recover the fixations from each trace using dispersion-threshold identification [11], which groups consecutive samples that stay within one degree of visual angle for at least 100 ms and replaces each group by its centroid. This yields a median of 13 fixations per observer.

Each observer's fixations, expressed in normalized image coordinates, are then turned into a probability density on a 64×64 grid by binning them and applying a Gaussian blur of one degree of visual angle, which approximates the region a single fixation samples at high acuity. The grid is deliberately coarser than the model input and kept fixed, since the quantities defined in Sec. V are entropies and would change with the discretization. Therefore, a common grid is makes them comparable across images. For the same reason each image is reduced to a common number of $N = 15$ observers.

V. Learning Framework

A. Network Architecture

The proposed network is an encoder-decoder structure. The encoder is a ResNet34 pre-trained on ImageNet, composed of five stages: the third, fourth and fifth layer are fed to the decoders in order to extract the multi-scale features. This is where it differs from the original DVA architecture, which uses a VGG16 encoder. Each decoder is a stack of upsampling blocks that double the resolution of their input and add the skip connection coming from the corresponding encoder stage, in a number sufficient to return to the 256×256 resolution of the input image. A Saliency Head, consisting of a 1×1 kernel and a sigmoid activation, turns each decoder output into an auxiliary saliency map, and a further Saliency Head fuses the three auxiliary maps into the final prediction, learning the fusion weights.

The auxiliary Saliency Heads consists in a 1×1 convolution followed by a sigmoid, while the Saliency Head that produces the final fused map is extended with two further refinements.

The first is a learned *center-bias prior*, which accounts for the well-known tendency of human fixations to concentrate near the center of the image regardless of its content. We implement it as a small 16×16 grid of learnable values, initialized to reproduce a centered Gaussian, which is upsampled to the resolution of the saliency map and added to it before the final sigmoid activation.

The second is a *Gaussian smoothing* applied to the map after the sigmoid activation. Since the ground-truth saliency maps of SALICON and MIT1003 are themselves obtained by blurring the fixation points, the predicted map is smoothed with a fixed 15×15 Gaussian kernel ($\sigma = 3$) in order to reduce the mismatch in spatial frequency between the predicted and the ground truth maps, before being rescaled by its maximum value to restore the $[0, 1]$ range.

From now on, the architecture just described is denoted as *Ours_Fusion*.

B. Loss Function

We formulate saliency prediction as a probability distribution estimation problem, where the objective is to predict the spatial probability distribution of human visual attention over an image. Under this formulation, we chose to use the Kullback-Leibler divergence (KLD) as loss function, combined with the Linear Correlation Coefficient (CC) score, in order to force the predicted saliency map $\mathcal{S}$ and the continuous saliency map $\mathcal{G}$ to be correlated. A similar approach is used in [6], where the authors use as loss function a combination of NSS, KLD, SIM and CC.

The loss used is obtained as a linear combination of KLD and CC as follows

$$\mathcal{L}(\mathcal{G},\mathcal{S}) = \lambda_1\mathcal{L}_{KL} + \lambda_2\mathcal{L}_{CC} \tag{1}$$

where $\lambda_1 = 1.0$ and $\lambda_2 = 0.5$.

The problem is treated as a deep supervised learning problem, so the training loss is computed over the final saliency map and the auxiliary saliency maps produced by the single decoders. Given M the number of the auxiliary maps, $\mathcal{F}$ the loss of the final saliency map and $\mathcal{A}_k$ the loss of the k-th auxiliary map, the final loss is computed as:

$$\mathcal{L}(\mathcal{G},\mathcal{S}) = \mathcal{F}(\mathcal{G},\mathcal{S}) + \frac{1}{M}\sum_{k=1}^{M}\mathcal{A}_k(\mathcal{G},\mathcal{S}) \tag{2}$$

C. Learning Framework

The network is trained with Adam, a weight decay of 10^{-4} and batches of 16 images. Two parameter groups set the initial learning rate to 10^{-3} for the decoder and 10^{-4} for the encoder, since the latter is pre-trained and should be fine-tuned more gently than the randomly initialized decoder. After a one-epoch linear warm-up both are decayed by cosine annealing down to 1% of their initial values; as the schedule acts as a multiplicative factor shared by the two groups, their $10\times$ ratio is preserved throughout. Training runs for at most 20 epochs and is stopped early if the validation loss does not improve by at least 10^{-4} over 7 consecutive epochs.

D. Observer Disagreement as a Target

Let g_i be the fixation density of observer i, $i = 1, \dots, N$, and $\bar{g} = \frac{1}{N}\sum_i g_i$ their average, which is what the aggregate map approximates. Defining $H(\cdot)$ as the Shannon entropy in bits, disagreement is measured by the Jensen-Shannon divergence across the N densities [12],

$$\mathrm{JS}\left(g_1, \dots, g_N\right) = H\left(\bar{g}\right) - \frac{1}{N}\sum_{i=1}^{N} H\left(g_i\right) \tag{3}$$

which is zero when all observers produce the same density and grows, up to $\log_2 N$, as they concentrate on disjoint regions.

Equation (3) cannot be used as a target on its own since part of the measured JS is affected by an upward bias [13].

We therefore estimate how large that term is by shuffling the observer labels, which is a standard correction for this bias [14]. All fixations of an image are pooled and redealt at random into synthetic observers of the original sizes, and (3) is recomputed and averaged over 200 redeals, giving $\mathrm{JS}_{\mathrm{null}}$. The shuffle keeps the same fixations and the same number of them per observer, and changes only who looked where. What remains after subtracting it,

$$\Delta_{\mathrm{JS}} = \mathrm{JS}_{\mathrm{obs}} - \mathrm{JS}_{\mathrm{null}} \tag{4}$$

that is the disagreement that noise alone cannot explain and it is the quantity the model predicts.

E. Agreement Head and Evaluation

The agreement head operates on the deepest encoder stage, which is reduced to a feature vector by global average pooling and mapped through two fully connected layers onto a mean μ and a log-variance $\log\sigma^2$. The features are obtained from the same forward pass that yields the saliency map, so the additional output introduces no computational overhead.

Encoder and decoders are frozen at this stage and held in evaluation mode, so that their batch normalization running statistics are not updated on MIT1003 and thus the density metrics reported in Sec. VI retain their status as a generalization test.

The head outputs a variance because Δ_{JS} is heteroscedastic: where observer behaviour is near-uniform, Δ_{JS} is a small quantity dominated by estimation noise. It is therefore trained with the β-weighted Gaussian negative log-likelihood of [15], with $\beta = 0.5$. The weighting prevents the model from minimizing the loss on poorly fitted samples by inflating their predicted variance.

Since the head is trained on MIT1003, which also serves as the test set of the saliency branch, it is evaluated by 5-fold cross-validation over images, reporting the Spearman correlation between the predicted and true Δ_{JS} pooled across the held-out folds. This correlation is interpreted against two reference values and is used to compute a bound of the correlation common to any model. The upper reference is

computable prior to training, and is used as a decision gate on whether the target admits prediction at all.

Finally, MIT1003 includes visually similar images acquired in the same capture sessions, and the folds are drawn at random over images without accounting for this grouping, so the reported correlation is expected to be marginally optimistic.

VI. Results

In this section, we present a quantitative evaluation of our approach using standard metrics for visual saliency prediction, comparing its performance against several state-of-the-art models on the MIT1003 and SALICON datasets.

A. Evaluation Metrics

We evaluate with five standard metrics: the Linear Correlation Coefficient (CC), Similarity (SIM), Kullback-Leibler Divergence (KLD), Normalized Scanpath Saliency (NSS) and AUC-Judd. The first three are distribution-based, treating both maps as probability distributions and measuring how well the overall density of human attention is reproduced, while the last two are location-based, scoring the predicted map at the discrete points where observers actually fixated. An orthogonal grouping separates the perception-based metrics (NSS, CC, SIM), which track alignment with human judgement, from the non-perception-based ones (KLD, AUC-Judd), more often used in task-specific applications such as target detection.

NSS and AUC-Judd both require discrete fixations, which we derive by thresholding the continuous ground-truth map, whereas published values are computed from the recorded fixation coordinates. Our values for these two metrics are therefore internally consistent but they are not directly comparable to performances reported elsewhere. Only CC, SIM and KLD are compared across settings in what follows.

B. Obtained Results

The results obtained by testing our network are compared against the results obtained by the DVA network of [1], which is the network that we aimed at improving, the MSI-Net of [5] and the TranSalNet_Res of [6] on the same test set. The quantitative results are shown in Tab. 1.

Model	CC ↑	SIM ↑	KLD ↓
DVA	0.64	0.50	-
MSI-Net	0.79	0.68	0.66
TranSalNet_Res	0.76	0.61	0.78
Ours_Fusion	**0.65**	**0.51**	**0.80**

TABLE 1: Comparison with other approaches on MIT1003.

Our model exceeds DVA on CC and SIM, but trails both MSI-Net and TranSalNet_Res on every distribution-based metric. The published values are obtained by fine-tuning each model on MIT1003, starting from SALICON-pretrained weights [6], whereas our network is trained on SALICON alone and evaluated on MIT1003 without any adaptation.

C. Ablation Study

In order to assess the effect of the individual design choices, we evaluated four variants of the proposed network on MIT1003 using the metrics described above. Each variant differs from its reference by a single factor, so that the resulting difference can be attributed unambiguously. The results are summarized in Tab. 2.

1) *Multi-decoder fusion*: *Ours_Res34* replaces the three decoder branches with a single decoder attached to the deepest encoder stage, keeping the ResNet34 backbone and the skip connections, so that the network reduces to a U-Net. Compared against *Ours_Fusion* it isolates the contribution of the fusion mechanism.
2) *Backbone depth*: *Ours_Res18* is the same single-decoder network on a ResNet18 backbone. Compared against *Ours_Res34* it isolates the effect of encoder capacity at a fixed decoder structure.
3) *Backbone family*: *Ours_PVT* keeps the three-branch structure but replaces the ResNet34 encoder with a pre-trained PVT v2 B0 [16], a transformer suited to image classification and allowing intermediate feature extraction. Compared against *Ours_Fusion* it isolates the effect of the encoder family.
4) *Loss function*: *Ours_MSE* is the single-decoder ResNet34 network trained with a mean squared error in place of (1). Compared against *Ours_Res34* it isolates the effect of the loss formulation.

The comparison shows that the loss dominates every architectural choice, instead the impact of architectural variations is several times smaller: the encoder is not the bottleneck and the additional parameter are not being exploited.

Taken together, the fusion mechanism does provide a small but consistent gain, visible on every metric, and is therefore retained in the proposed model. The gap that remains with respect to the heavier state-of-the-art models of Tab. 1 is however unlikely to be closed by further architectural changes at this scale: the ablation points at the objective as the component with the most leverage, which is consistent with the fact that the best performing model we compare against [6] combines four complementary saliency metrics rather than two.

Model	CC ↑	SIM ↑	KLD ↓	NSS ↑	AUC-Judd ↑
Ours_Res34	0.643	0.504	0.832	2.096	0.924
Ours_Res18	0.648	0.510	0.810	2.121	0.928
Ours_PVT	0.643	0.510	0.824	2.107	0.924
Ours_MSE	0.606	0.476	0.918	1.983	0.911
Ours_Fusion	**0.650**	**0.512**	**0.801**	**2.145**	**0.928**

TABLE 2: Ablation study on MIT1003.

D. Observer Disagreement

We now turn to the second output of the model: the saliency branch is frozen at this stage, so what follows concerns only the predicted disagreement.

Before training any model, the target itself is worth inspecting. Of the raw divergence $\mathrm{JS}_{\mathrm{obs}} = 1.140$ bits averaged over MIT1003, the permutation null accounts for $\mathrm{JS}_{\mathrm{null}} = 0.808$ bits, i.e. 71% of it is sampling noise rather than real disagreement – confirming that the correction of (4) is doing substantive work. The remaining $\Delta_{\mathrm{JS}} = 0.332$ bits is positive for 99.8% of the images.

Its split-half reliability is $\rho_{tt} = 0.473$, which caps the correlation any model can achieve at $\sqrt{\rho_{tt}} \approx 0.688$: this is the ceiling the results below should be read against.

Model	per-fold ρ	pooled ρ
Ours_Fusion	0.514	0.503
Ours_PVT	0.512	0.509
Ours_Res34	0.532	0.525
Three-feature baseline	–	−0.001
Attainable maximum	–	0.688

TABLE 3: Prediction of observer disagreement on MIT1003.

All three architectures predict disagreement well above chance (Tab. 3), the best being *Ours_Res34* at pooled $\rho = 0.525$, about 76% of the ceiling. The ridge regression on edge density, colour spread and face area scores $\rho = -0.001$: whatever drives disagreement is not captured by these low-level statistics, yet it is recoverable from encoder features trained only to predict where people look, on a different dataset.

The architectures, however, are indistinguishable from one another: they fall within 0.022 pooled ρ, inside the spread across folds, with the simplest model ranking first. This contradicts our expectation that self-attention would help on a target driven by competition between distant regions, and mirrors the tie already seen on the standard metrics of Tab. 2.

VII. Concluding Remarks

We redesigned the Deep Visual Attention Prediction architecture with a ResNet34 encoder, U-Net-style skip connections and a joint Kullback-Leibler and correlation loss. Moreover, we introduced a second target, inter-observer disagreement, and computed from the individual eye-tracking records that saliency pipelines discard. On MIT1003, and without any fine-tuning on that dataset, the model exceeds DVA on CC and SIM though it does not reach the distribution metrics of the heaviest architectures.

Two findings stand out. The objective matters far more than the architecture: replacing the joint loss with a mean squared error costs about five times more CC than removing the multi-decoder fusion, and a deeper encoder brings no gain. Disagreement is also predictable, reaching 76% of the attainable correlation against a baseline of low-level image statistics that performs at chance, but all three architectures predict it equally well: self-attention brings no advantage, contrary to our expectation that it would help on a quantity driven by competition between distant regions. No target separate the three architectures, and the main limitations of this study follow from the same source: a single test set and a dataset, MIT1003, too small to push further. Extending both to larger corpora is the natural next step. In the meantime, since the second head costs virtually nothing to add, it already provides a cheap way to asses how representative a predicted map is likely to be of any single viewer.

Lesson Learned and Challenges

Access to GPUs through Colab was limited by daily quotas, which repeatedly bottlenecked training. The most instructive difficulty was that the MIT1003 distribution stores raw gaze traces rather than fixations, and our first loader treated every sample as a fixation. Nothing failed and the numbers looked plausible; only checking the fixation count per observer revealed they were meaningless. Beyond this, the project let us combine architectures and losses from the course in ways none of them were designed for, alongside material from other courses, in particular Computer Vision.

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